

ARM (STM32F401) Graduation Project

Portable Health Monitoring Device with Wireless Data Transmission

Requirements Analysis Document

Team Members	Omar, Khaled, Hussien
Project Duration	2 weeks (14 days) to submission
Selected Project	Project 2 — Portable Health Monitoring Device with Wireless Data Transmission
Document Status	Draft v1.0

1. Introduction

1.1 Purpose

This document defines the functional and non-functional requirements for the Portable Health Monitoring Device with Wireless Data Transmission and records the task breakdown agreed by the team for the two-week implementation window. It serves as the shared reference for scope, ownership, and delivery milestones.

1.2 Project Overview

The device is a portable health monitor built on the STM32F401 platform. It measures heart rate and physical activity, presents the results through an on-device dashboard, and transmits health data wirelessly for remote monitoring. The system is a direct implementation of the "Project 2" specification from the ARM training graduation brief.

1.3 Objectives

- Continuously monitor heart rate and physical activity with on-device feedback.
- Transmit health data securely to a cloud service or companion app in real time.
- Provide an intuitive interface (dashboard, settings, history) for the end user.
- Alert the user audibly and visually to critical conditions or low battery.
- Deliver a working, integrated, and tested prototype within two weeks.

2. Scope

In scope: heart-rate sensing, step/activity tracking, Wi-Fi based cloud/app connectivity, the on-device UI (TFT, 7-segment, LED matrix), audio alerts, power management, and manual button controls — as described in Functional Requirements FR-1 through FR-7 below.

Out of scope: mobile/companion app development beyond a receiving endpoint for transmitted data, mechanical enclosure design, and multi-user/account management.

3. Functional Requirements

Each requirement below corresponds to one functional block of the device and is assigned a single owner responsible for both the low-level driver and the feature logic built on top of it.

ID	Requirement	Key Hardware	Priority	Owner
FR-1	Heart Rate Monitoring — continuously measure heart rate, display on TFT, log data for trend review, trigger alerts when readings fall outside a safe range.	Heart rate sensor	Must-have	Omar
FR-2	Step Counter & Activity Tracking — count steps and calculate	Accelerometer	Must-have	Khaled

ID	Requirement	Key Hardware	Priority	Owner
	distance from accelerometer data; distinguish walking, running, and inactivity; show steps, distance, and calories on the TFT.			
FR-3	Wireless Health Data Transmission — connect to a cloud service / companion app over Wi-Fi; sync health data in real time using a secure (TLS) channel.	ESP Wi-Fi module	Must-have	Hussien
FR-4	Interactive User Interface — dashboard view (heart rate, steps, battery, connectivity), settings menu for thresholds/preferences, historical graphs of heart rate and activity.	TFT display, 7-segment displays	Must-have	Omar
FR-5	Audio Feedback & Alerts — play audio alerts for critical conditions (abnormal heart rate, low battery) and instructional/guided audio for setup and device use.	Audio jack, R2R DAC	Should-have	Hussien
FR-6	Power Management & Manual Controls — display battery status and low-battery alerts, enter a power-saving mode when idle, and provide button controls for power/reset/menu navigation.	Buttons, LEDs	Must-have	Hussien
FR-7	LED Matrix Data Visualization — show status animations (syncing, alert active) and simple visualizations of activity level or heart-rate status.	LED matrix, serial-to-parallel ICs	Should-have	Khaled

4. Non-Functional Requirements

Category	Requirement
Security	All health data transmitted over Wi-Fi must use a secure protocol (TLS); no health data is sent in plaintext.
Reliability	The device must recover gracefully from a lost Wi-Fi connection and continue local monitoring/logging while disconnected.
Performance	Heart rate and activity readings must update on the dashboard with no perceptible lag (target: under 1 second per update cycle).
Usability	Core status (heart rate, steps, battery, connectivity) must be readable at a glance on the dashboard without navigating menus.

Power efficiency	The device must enter a low-power mode automatically after a defined idle period to preserve battery life.
Maintainability	Firmware follows a layered architecture (driver / feature-logic separation) so each module can be developed, tested, and reviewed independently.

5. Team Roles & Task Assignment

Ownership is assigned so that whoever implements a module's low-level driver also implements the feature logic on top of it, avoiding hand-off gaps between driver and application code. Khaled has specifically requested ownership of the accelerometer / activity-tracking module.

Member	Owned Modules (driver + feature logic)	Responsibilities
Omar	FR-1 Heart Rate Monitoring FR-4 Interactive User Interface	Heart rate sensor driver + real-time read/alert logic + data logging. TFT/7-segment driver, dashboard, settings menu, and historical graph rendering. Defines the shared health-data structure used by all modules.
Khaled	FR-2 Step Counter & Activity Tracking FR-7 LED Matrix Data Visualization	Accelerometer driver + step-counting and activity-recognition logic (walk/run/idle), distance & calorie calculation. LED matrix driver and status/activity animations.
Hussien	FR-3 Wireless Health Data Transmission FR-5 Audio Feedback & Alerts FR-6 Power Management & Manual Controls	ESP Wi-Fi driver + cloud/app sync logic with TLS. Audio jack/R2R DAC driver + alert and guided-audio playback. Button driver + battery monitoring and low-power mode logic.

Workload is split 2 / 2 / 3 across Omar, Khaled, and Hussien respectively; Hussien's three modules (Wi-Fi, audio, power/buttons) are individually lighter in scope than the sensor and UI modules, keeping total effort balanced across the team.

6. Project Timeline (2 Weeks)

Days	Milestone
Days 1–2	Kickoff: agree on the shared data structure and module interfaces (heart-rate value, step count, activity state, battery %, alert flags, connectivity status) so all three members can build in parallel.
Days 3–8	Parallel development: each member implements their owned driver(s) and feature logic against the agreed interface. Short daily sync.
Days 9–11	Integration: wire sensor data into the UI, alerts into audio/LED outputs, and health data into the Wi-Fi upload path. Resolve interface mismatches.
Days 12–13	End-to-end testing: edge cases such as low battery, Wi-Fi disconnection, noisy sensor readings, and alert thresholds.
Day 14	Polish, full demo run-through, and finalize documentation/slides for submission.

7. Assumptions & Constraints

- Heart rate sensor and accelerometer are additional hardware and are assumed to be available to the team before Day 1.
- A cloud endpoint or simple app/backend for receiving transmitted health data is available or will be mocked for demo purposes.
- All three members have access to their own STM32F401-based hardware set for parallel development.
- The shared data structure agreed on Days 1–2 is treated as a fixed interface for the rest of the sprint to avoid integration delays.

8. Deliverables & Acceptance Criteria

- Working firmware implementing FR-1 through FR-7 on the target hardware.
- Live demo showing heart-rate and activity monitoring, dashboard UI, audio/LED alerts, and wireless data sync.
- Source code organized by module/owner in the team's repository.
- Short project report covering design decisions, task ownership, and known limitations.